TN TEACHERS BENCHMARK PORTAL

OFFICIAL TNTET EXAM LEARNING & MATERIALS HUB • TNTEACHERS.IN

PRODUCTION V2.5
ENTERPRISE
Live URL: https://tnteachers.in

Project Delivery, Module Specification & Technical Architecture

Executive technical handover document prepared for client presentation, system verification, and final payment release.

1. Executive Summary & Delivery Verification

The **TN Teachers Portal (tnteachers.in)** has been successfully engineered, cloud-deployed, and verified on high-availability Linux infrastructure. The platform serves Tamil Nadu Teacher Eligibility Test (TNTET) Paper 1 and Paper 2 candidates, school educators, and competitive scholarship students with comprehensive learning workflows, automated examinations, and textbook repositories.

9 Core Modules DELIVERED
3 Premium Exam Preparation Tracks + 6 Free Public Curriculum Modules with real-time sync.

Razorpay Integration ACTIVE
End-to-end checkout, lead registration, combo course activation, and admin manual override.

PWA & Mobile-First ACTIVE
Installable web app on Android & iOS with offline caching, fast touch manipulation, and safe areas.

2. Detailed 9-Module System Specification

A. Premium Locked Modules (Combo Access Pass)

Monetization Rule: Enrolling or paying for either TET 1 or TET 2 automatically unlocks all 3 modules simultaneously. Until enrolled, these modules display a lock icon, status badge, and open the enrollment modal.

1. TET – 1 (Paper 1: Classes 1 to 8) PREMIUM LOCKED

Designed for Secondary Grade Teacher (SGT) candidates. Includes structured chapter-by-chapter curriculum summaries, integrated video lessons with fullscreen playback, interactive practice MCQs with pedagogical theory explanations, and a timed 80% qualifying benchmark exam with automated scoring.

2. TET – 2 (Paper 2: Classes 6 to 12) PREMIUM LOCKED

Designed for Graduate Teacher (BT Assistant) candidates across Science, Mathematics, Social Science, Tamil, English, and Child Development & Pedagogy (CDP). Features high-school and higher-secondary chapters, dual question banks (Type-1 Primary & Type-2 Alternate) to ensure authentic testing without memorization.

3. Online Live Class & Meeting Portal PREMIUM LOCKED

Live interactive video classroom integrating with Google Meet, Zoom, YouTube Live, and Webex. Features a real-time pulsing **LIVE NOW** notification banner when class is active, one-click direct access for enrolled students, and upcoming class countdown schedules.

B. Free Public Educational Modules

4. TN Textbooks Explorer

FREE PUBLIC

Complete Class 1 to 12 curriculum repository. Multi-filter by Class, Subject, Medium (Tamil/English), and Term. Features an in-browser PDF reader with Admin download toggles.

5. Minnal News (மின்னல் செய்திகள்)

FREE PUBLIC

Opens as a dedicated mobile feed (`/news`) with real-time flash educational updates on TRB hall tickets, exam schedules, and School Education Government Orders (G.O.).

6. Tamil Reading (தமிழ் வாசிப்பு)

FREE PUBLIC

Language enrichment module with pronunciation guides, grammatical literature (இலக்கணம்/ இலக்கியம்), and comprehension exercises for teacher candidates.

7. English Reading Mastery

FREE PUBLIC

Grammar drills, phonetics, comprehension passages, and pedagogical English vocabulary builder tailored for TNTET language proficiency.

8. NMMS Exam Preparation

FREE PUBLIC

National Means-cum-Merit Scholarship portal for Class 8 students with Mental Ability Test (MAT) and Scholastic Aptitude Test (SAT) practice questions and model tests.

9. TREAST / TRUST Exam Hub

FREE PUBLIC

Tamil Nadu Rural Students Talent Search Examination portal offering subject-wise practice tests and model question papers for government school students.

3. Razorpay Monetization & Student Enrollment System

A dedicated commercial checkout and enrollment engine has been integrated to allow seamless monetization of teacher eligibility prep courses:

> 💰 **End-to-End Course Purchase & Activation Flow:**
>
> 1. **Pre-Payment Lead Capture:** Clicking any locked module opens the Enrollment Modal collecting candidate's **Full Name**, **10-digit Mobile Number** (+91 validation), **Email Address**, and **Designation** (Teacher / Aspirant / B.Ed / D.T.Ed / Other).
> 2. **Database Recording:** Inquiries are instantly recorded in the production MySQL `enrollments` table as pending leads before checkout.
> 3. **Razorpay Checkout:** Dynamically loads the official Razorpay JS gateway with customer info pre-filled and course amount (configurable in Admin Settings).
> 4. **HMAC Signature Verification:** Backend verifies the payment signature securely using the secret key before activating course access.
> 5. **All-In-One Combo Pass:** Enrolling or paying for either TET 1 or TET 2 unlocks all 3 modules simultaneously (TET 1 + TET 2 + Online Live Class).
> 6. **Admin Free Enrollment Override:** Administrators can manually enroll any candidate for free from the Admin Panel without requiring a payment gateway transaction.

4. Super-Admin Management Center (`/admin`)

ADMIN MANAGEMENT TOOL	CAPABILITIES & FUNCTIONAL ROLE
Student Enrollments CRM	Search, filter, and monitor all student inquiries, transactions, and payment IDs. Includes `+ Enroll Candidate Free` button to grant instant free access.
PDF Download Controls	Admin Settings toggles to instantly enable or disable student Material PDF downloads and Quiz PDF downloads across the entire website.
Curriculum & Lesson CMS	Upload textbook PDFs, configure notes, attach YouTube video embeds, and organize content across classes 1 to 12.
AI Question Bank Engine	Manage Type 1 and Type 2 question banks with Google Gemini AI question generator, single-question regenerator, and offline PDF test export.
Live Class Scheduler	Schedule live sessions with Google Meet/Zoom links, monitor active status, and broadcast real-time "LIVE NOW" alerts to candidates.
Daily Practice CMS	Create daily class-wise timed tests with custom stage configurations (Notes, Video, Practice, Quiz).
Announcements Ticker	Publish real-time breaking ticker notices displayed across the candidate dashboard with priority tags.
Dashboard Containers CMS	Configure, reorder, and customize badges/titles for all 9 dashboard learning containers.

5. Progressive Web App (PWA) & Mobile UX

Progressive Web App (PWA)

Equipped with `manifest.json`, `sw.js` Service Worker, 192px and 512px maskable icons. Enables users to install the app on Android and iOS home screens with offline caching support.

Deep-Link Auth Redirection

Visiting specific lesson links (e.g., `/learning/lesson/3?stage=2`) prompts sign-in immediately, then redirects directly to that exact lesson upon authentication. Normal student logins redirect strictly to `/student`.

6. Technical Stack Specifications

COMPONENT	TECHNOLOGY & VERSION	PURPOSE & SPECIFICATION
Frontend Framework	React 19.2 + Vite 8.2	Component-based SPA with fast ESM bundling and minimal bundle footprint.
Styling & Design	Tailwind CSS v4 + Lucide React	Utility-first CSS design system with responsive mobile layouts and SVG icons.
Backend API	FastAPI 0.110 (Python 3.10/3.13)	High-performance asynchronous REST API framework with Pydantic v2 schemas.
ASGI Server	Uvicorn 0.28	Production event-loop server for async concurrency and high throughput.
ORM & Database	SQLAlchemy 2.0 + MySQL 8.0	Enterprise relational ORM connected to MySQL `tet_platform` database.
Cloud Infrastructure	Ubuntu 22.04 LTS VPS (66.116.250.237)	Dedicated server with systemd process management and Nginx reverse proxy.
SSL & Security	Let's Encrypt Certbot SSL	HTTPS encryption with auto-renewing certificate and HMAC payment verification.

7. Integrated APIs (External & Internal)

A. External Cloud APIs & SDKs

- **Razorpay Payment Gateway API:** Client SDK (`checkout.js`) and server-side HMAC-SHA256 signature verification.
- **Google OAuth 2.0 / Identity Services API:** Client token verification for 1-click Google Sign-In.
- **Google Gemini Generative AI API:** AI-powered contextual question generation and question replacement from textbook text.
- **YouTube Player Embed API:** Integrated responsive classroom lecture playback across all lessons.

B. Key Internal REST API Endpoints

METHOD	ENDPOINT	FUNCTIONAL DESCRIPTION
`POST`	`/api/enrollments/register`	Capture candidate lead details and initiate course registration.
`POST`	`/api/enrollments/verify-payment`	Verify Razorpay payment signature and unlock TET-1, TET-2, and Live Class.
`GET`	`/api/enrollments/my-status`	Verify candidate enrollment status and active course access permissions.
`POST`	`/api/admin/enrollments/manual`	Admin override to grant 100% free lifetime access to any student.
`GET`	`/api/public-settings`	Public endpoint returning download permission toggles and Razorpay key.

METHOD	ENDPOINT	FUNCTIONAL DESCRIPTION
`GET`	`/api/learning/lesson/{id}`	Fetch lesson details, study notes, video embed, and quiz info.
`POST`	`/api/learning/lesson/{id}/submit-test`	Evaluate student quiz answers, score percentage, and issue pass/fail result.
`GET`	`/api/meetings/active`	Fetch active live classes (triggers "LIVE NOW" pulse badge).
`GET`	`/api/materials`	Filter textbook PDFs by Class (1–12), Subject, Medium, and Term.
`GET`	`/api/news/live`	Retrieve real-time educational flash news and government orders.

💼 Final Milestone Handover & Payment Request

The complete platform implementation, database migration, Razorpay monetization engine, PWA installation setup, and production deployment on **https://tnteachers.in** have been finalized and verified.

Project Title: Tamil Nadu Teachers Eligibility Benchmark Portal
Live Web Portal: https://tnteachers.in
Repository: GitHub / tho-mas12 / tet-color
Milestone Deliverable: Full Platform Delivery, CMS, Monetization & Cloud Deployment
Status: Approved & Production Live